Subrat Prasad Panda

Ph.D. Candidate, Computer Science, NTU Singapore

Trustworthy RL · Neurosymbolic AI

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Research Statement

I am broadly interested in **Trustworthy Reinforcement Learning**, particularly in designing interpretable, robust, and verifiable policies for safety-critical domains. My current work focuses on **Neurosymbolic** approaches such as programmatic policies, combining explicit knowledge with learning to bridge System 1 (neural) and System 2 (symbolic) reasoning.

Research Areas and Skills

Reinforcement Learning	Hierarchical and goal-conditioned RL, sample-efficient learning in sparse-reward and long-horizon tasks, robust and distributional RL, RL fine-tuning of foundation models
Neurosymbolic AI	Programmatic and interpretable policies (differentiable decision trees), symbolic planning with incremental knowledge, interpretable state abstractions and world models
Safety & Verification	Formal verification of learned controllers, reach-avoid and safety certificates, abstraction-based reasoning for safety-critical systems
Applications	Cyber-physical systems, microgrid energy management, autonomous navigation and robotics, LLM-based planning for embodied agents
Programming	Python, C, PyTorch, TensorFlow, ROS 2, MATLAB, LaTeX
Coursework	Mathematical Statistics, Deep Learning, Machine Learning, Computer Vision

Education

2023 – 2027 expected — **Ph.D., Computer Science** — Singapore
Nanyang Technological University (NTU)
Thesis: *Neurosymbolic Techniques for Safe and Efficient Reinforcement Learning*
Advisors: Arvind Easwaran (NTU), Blaise Genest (IPAL, CNRS France & Singapore)

2019 – 2021 — **M.Tech., Computer Science** — Kolkata, India
Indian Statistical Institute (ISI)
Thesis: *Strategies for User Allocation and Service Placement in Multi-Access Edge Computing*
Advisor: Ansuman Banerjee CPI: 82.1%

2012 – 2016 — **B.Tech., Electronics and Instrumentation** — Silchar, India
National Institute of Technology (NIT) Silchar
Thesis: *Real-Time Position Control of DC Motor using PID and Fuzzy Logic*
Advisor: Munmun Khanra CPI: 7.92/10

Research Experience

2022 – 2023 — **Research Associate** — Singapore
CNRS@CREATE
- Designed a granular control system for microgrids using Deep Reinforcement Learning.

2021 – 2022 — **Machine Learning Engineer** — Pune, India
InFocusp Innovation Pvt. Ltd., under attachment to Google X (USA)
- Developed solutions for the Abstraction and Reasoning Challenge (ARC) using Reinforcement Learning.

2021 – 2021 — **Research Engineer** — Hyderabad, India
TiHAN, IIT Hyderabad
- Developed braking and throttle control systems using Arduino for a test vehicle.
- Deployed a lane detection model on the NVIDIA AGX board.

Publications

Journal Articles

1. **S. P. Panda**, B. Genest, and A. Easwaran, **Approximation-Free Differentiable Oblique Decision Trees**, *Journal of Machine Learning Research (JMLR)*, vol. 27, 2026.
2. **S. P. Panda**, B. Genest, A. Easwaran, R. Rigo-Mariani, and P. Lin, **Methods for Mitigating Uncertainty in Real-Time Operations of a Connected Microgrid**, *Sustainable Energy, Grids and Networks*, vol. 38, p. 101 334, 2024.

Conference Proceedings

1. **S. P. Panda**, B. Genest, and A. Easwaran, **Neurosymbolic Reasoning with Incremental Knowledge for Sample Efficient Hierarchical Reinforcement Learning**, in *ECML-PKDD*, 2026.
2. **S. P. Panda**, B. Genest, A. Easwaran, and P. N. Suganthan, **Vanilla Gradient Descent for Oblique Decision Trees**, in *European Conference on Artificial Intelligence (ECAI)*, IOS Press, 2024.
3. **S. P. Panda**, A. Banerjee, and A. Bhattacharya, **User Allocation in Mobile Edge Computing: A Deep Reinforcement Learning Approach**, in *2021 IEEE International Conference on Web Services (ICWS)*, 2021, pp. 447–458.
4. **S. P. Panda**, K. Ray, and A. Banerjee, **Service Allocation/Placement in Multi-Access Edge Computing with Workload Fluctuations**, in *International Conference on Service-Oriented Computing*, 2021, pp. 747–755.
5. **S. P. Panda**, K. Ray, and A. Banerjee, **Dynamic Edge User Allocation with User Specified QoS Preferences**, in *International Conference on Service-Oriented Computing*, 2020, pp. 187–197.

Under Review

1. **S. P. Panda**, B. Genest, and A. Easwaran, **Neurosymbolic Learning of Interpretable Abstraction in Reinforcement Learning**, AAAI Student Abstract (submitted), 2027.
2. **S. P. Panda**, M. Prashant, and A. Easwaran, **CriOC: Criticality-Balanced Option-Critic via Distributional RL**, in *AAMAS (submitted)*, 2027.
3. **S. P. Panda**, L. Yuke, B. Genest, D. Zikelic, and A. Easwaran, **Learning Oblique Abstractions for the Safety Verification of Oblique DT Policies**, in *ICLR (submitted)*, 2027.

Workshops and Talks

1. **S. P. Panda**, **Neurosymbolic Techniques for Safe Reinforcement Learning**, Invited talk, 68NQRT Seminar, ICUR, Rennes, France, Sep. 2026.
2. **S. P. Panda**, **Reinforcement Learning**, Tutorial lecture with hands-on notebooks, Deep Learning Bootcamp 2026, NTU Singapore, 2026.
3. **S. P. Panda**, B. Genest, and A. Easwaran, **Programmatic Reinforcement Learning for Trustworthy Microgrid Management**, AI4X International Conference, 2025.

Open-Source Software and Educational Content

Research Code

DTSemNet Differentiable oblique decision trees and tree policies for RL (JMLR 2026, ECAI 2024).

InK-BWTS Neurosymbolic hierarchical RL with Belief World Tree Search (ECML-PKDD 2026).

F1Tenth ROS 2 race stack (SLAM, particle-filter localisation, Pure Pursuit, etc.) on Jetson AGX.

Educational Content (ongoing effort)

Deep RL Jutsu Notebook-style tutorials implementing deep RL algorithms from scratch (DQN, DDPG, SAC, PPO), with a curated guide to RL libraries and resources.

GenAI Small-scale implementations of generative models (VAE, GAN, diffusion, state-space models, CLIP, VLA) on toy problems to explain how they work.

RLFT Tutorial series on RL fine-tuning of foundation models: RLHF, DPO and GRPO, from a from-scratch GPT to a pretrained LLM.

Awards and Scholarships

2023 – 2027	**DesCartes Scholarship**: Ph.D. scholarship awarded by CNRS@CREATE under the DesCartes project (research collaboration between Singapore and France).
2021	**M.Tech Best Thesis Nomination**: ranked among the top 5 theses of the batch.
2019 – 2021	**M.Tech Scholarship**: scholarship awarded to pursue a master's degree.
2006 – 2012	**Jawahar Navodaya Vidyalaya (JNV) Scholarship**: fully funded residential education from Class 6 to 12.

Academic Services

Teaching and Mentoring

Final Year Projects — **Undergraduate Final Year Projects** — NTU

Project supervision

SafeDAG: Precedence-Constrained LLM Steering for Safer Planning in Embodied Agents (ongoing); programmatic policies for interpretable RL using pre-trained models / LLMs; robustness vs. explainability trade-offs in CNNs.

URECA — **Undergraduate Research Experience on Campus** — NTU

Research mentoring

Mechanistic interpretability of Decision Transformers; ARC reasoning with LLMs; physics-inspired model-based RL; residual reinforcement learning.

F1Tenth — **F1Tenth Autonomous Racing Team** — NTU

Mentor

Built the ROS 2 race stack on Jetson AGX and advised students on control and navigation strategies.

Reviewing

Reviewer — CAV, AAAI, ECAI, IEEE TSMC, RTSS, NeurIPS, AAMAS, NeSy.

Leadership

Vice President — **CCDS Student Club**, NTU: organized events such as the Deep Learning Bootcamp and AI Meetup.

Academic References

Ph.D. Advisor	**Arvind Easwaran**, Associate Professor, CCDS, NTU Singapore	arvinde@ntu.edu.sg
Ph.D. Advisor	**Blaise Genest**, Research Scientist, CNRS@CREATE, Singapore	blaise.genest@cnrsatcreate.sg
M.Tech Advisor	**Ansuman Banerjee**, Professor, ISI Kolkata, India	ansuman@isical.ac.in

Academic Collaborators

Verification	**Đorđe Žikelić**, Assistant Professor, CCDS, NTU Singapore	djordje.zikelic@ntu.edu.sg
Decision Trees	**P. N. Suganthan**, Research Professor, KINDI, Qatar University	p.n.suganthan@qu.edu.qa
Edge AI	**Arani Bhattacharya**, Assistant Professor, IIIT Delhi, India	arani@iiitd.ac.in